The Non-Abelian Kelvin–Helmholtz Instability

A shear-driven SU(2) plasma instability: theory and GPU validation

SU(2) Yang–Mills KH simulation program

2026-07-22

Motivation

Two counter-streaming, color-charged plasma beams with a shear flow layer between them, coupled to a non-Abelian SU(2) gauge field.

- Classical (Abelian) Kelvin–Helmholtz instability is textbook physics for neutral shear flows. Its non-Abelian analogue is not.
- Relevant wherever colored plasma has shear: early-universe electroweak plasma, quark–gluon plasma, and proposed dark non-Abelian sectors.
- **No prior work** treats a genuinely shear-driven, non-Abelian instability with a full wavenumber map. The closest results in the literature are either two-stream-driven (Manuel–Mrówczyński) or anisotropy-driven chromo-Weibel modes (Mrówczyński et al.) — not shear.

Physical picture

This talk derives the instability from first principles, finds an exact reduction to a single ODE, extracts a growth-rate ceiling and a wavelength-selection rule, and confronts all of it with GPU simulation data.

The physical system, before code-unit normalization

Same fields, in Gaussian (cgs) units with charge, mass, and c restored; g is the bare Yang–Mills coupling. Reconstructed from the code's own normalization recipe ($c \to 1$, background charge $Q_0 \to 1$, $4\pi n_0 \to 1$, $g \to \alpha$), not pulled from a separately published source — sanity-check before use.

$$\partial_t n + \nabla\cdot(n\vec{v}) = 0, \qquad \partial_t p_x + \nabla\cdot(\vec{v}p_x) = -n\textstyle\sum_a Q^a\left(E_x^a + \frac{v_z}{c}B_y^a\right)$$

$$\partial_t p_z + \nabla\cdot(\vec{v}p_z) = -n\textstyle\sum_a Q^a\left(E_z^a - \frac{v_x}{c}B_y^a\right), \qquad \partial_t Q^a = g\,\frac{v_z}{c}(\vec{Q}\times\vec{A}_z)^a$$

$$\partial_t E_x^a = -c\,\partial_z B_y^a - gc(\vec{A}_z\times\vec{B}_y)^a - 4\pi J_x^a, \qquad \partial_t E_z^a = c\,\partial_x B_y^a - 4\pi J_z^a$$

$$\partial_t B_y^a = c\,\partial_x E_z^a - c\,\partial_z E_x^a + gc(\vec{A}_z\times\vec{E}_x)^a, \qquad \partial_t A_z^a = -c\,E_z^a$$

$\vec{v} = \vec{p}/(mn)$; $J_x^a = -\left[n_A Q_A^a v_{xA} + n_B Q_B^a v_{xB}\right]$ (likewise J_z^a). Setting $c = 1$, $|Q_A| = |Q_B| = Q_0 \to 1$, $4\pi n_0 \to 1$, and writing $\alpha \equiv g$ recovers the code-unit system of the next slide exactly.

The simulated system

For each color $a \in \{1, 2, 3\}$: fields $E_x^a, E_z^a, B_y^a, A_z^a$. For each beam $s \in \{A, B\}$: fluid density n_s, momenta p_{xs}, p_{zs}, color charge Q_s^a.

$$\partial_t n + \nabla \cdot (n\vec{v}) = 0, \qquad \partial_t p_x + \nabla \cdot (\vec{v} p_x) = -n \textstyle\sum_a Q^a (E_x^a + v_z B_y^a)$$

$$\partial_t p_z + \nabla \cdot (\vec{v} p_z) = -n \textstyle\sum_a Q^a (E_z^a - v_x B_y^a), \qquad \partial_t Q^a = \alpha\, v_z (\vec{Q} \times \vec{A}_z)^a$$

$$\partial_t E_x^a = -\partial_z B_y^a - \alpha (\vec{A}_z \times \vec{B}_y)^a - J_x^a, \qquad \partial_t E_z^a = \partial_x B_y^a - J_z^a$$

$$\partial_t B_y^a = \partial_x E_z^a - \partial_z E_x^a + \alpha (\vec{A}_z \times \vec{E}_x)^a, \qquad \partial_t A_z^a = -E_z^a$$

Beam A: $v_z = +V_0 \tanh \xi$, $Q^1 = +1$; beam B: $v_z = -V_0 \tanh \xi$, $Q^1 = -1$; background potential $A_{z1}(\xi) = -V_0 \ln \cosh \xi$ sets $u(\xi) \equiv -\alpha A_{z1}(\xi) \geq 0$. α is the non-Abelian coupling; V_0 the shear velocity; k_z the perturbation wavenumber along the flow.

The six-field linear system

Linearizing about the sheared background in the color-2/3 sector (which carries charge ± 1 under the residual color-1 axis) gives a closed system of six complex fields, $b, e_x, e_z, a, q_A, q_B \propto e^{\mathrm{i}k_z z + \gamma t}$:

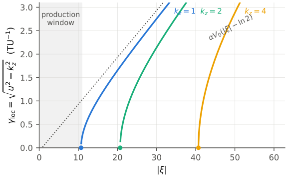
$$\boxed{\begin{aligned} \gamma\, b &= -\mathrm{i}\,\Omega_F\, e_x + e_z' & \gamma\, a &= -e_z \\ \gamma\, e_x &= -\mathrm{i}\,\Omega_A\, b & \gamma\, q_A &= \mathrm{i}\alpha v\, a - \mathrm{i} v\, \Omega_A\, q_A \\ \gamma\, e_z &= b' + v\,(q_A - q_B) & \gamma\, q_B &= \mathrm{i}\alpha v\, a + \mathrm{i} v\, \Omega_A\, q_B \end{aligned}} \qquad \Omega_A = k_z - u(\xi), \quad \Omega_F = k_z + u(\xi)$$

Validated directly against the 2D simulation: 1–4% agreement on the core validation series, 4–9% median across the full wavenumber map.

Two branches emerge from the same background

Freezing the background locally, the system splits into an electromagnetic block and a charge-precession block.

$$\underbrace{\gamma^2 = -\Omega_A\Omega_F = u^2 - k_z^2}_{\text{outer, tachyonic branch}} \qquad \underbrace{\gamma^2 = -g(\xi;\gamma), \quad g = \frac{2\alpha v^3\Omega_A}{\gamma^2 + v^2\Omega_A^2}}_{\text{shear branch, inside the layer}}$$



Physical picture

Far from the layer, the strong background alone is unstable — a Nielsen–Olesen-type charged-vector mode, independent of the flow. Inside the layer, the flow shakes the beams' color charge; their *difference* current feeds back into the field — a genuine shear-driven resonance. This talk focuses on the shear branch.

The $k_z = 0$ limit: a validated anchor

At $k_z = 0$ the shear-branch dispersion relation reduces to a cubic with an exact closed-form growth rate:

$$\boxed{\gamma(k_z{=}0) = \left(\sqrt{\frac{\alpha^3}{2}}\, V_0\right)^{1/3} \sin 60^\circ}$$

Physical picture

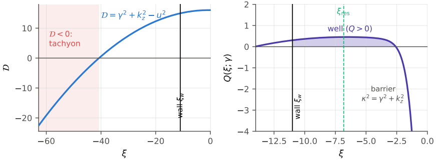
Not a bunching instability — nothing varies along z. It is *filamentation*: the background field rotates color-2/3 fields into each other while the beams' counter-streaming currents feed the growth. This is a member of the well-known chromo-Weibel family, appearing here as the $k_z \to 0$ limit of the shear branch.

Validated to 0.5%: at $\alpha = 2$, $V_0 = 0.1$, simulated $\gamma = 0.5039$ vs. predicted 0.5065, over nine decades of growth — this is the strongest, parameter-free anchor for everything that follows, and it holds on two independent simulation configurations three weeks apart.

Reduction to a single ODE

The six-field system has only one spatial derivative chain; every other field is algebraic in $a(x)$, the color-2/3 potential perturbation. Eliminating them gives one second-order equation:

$$\boxed{\left(\frac{\gamma^2 a'}{\mathcal{D}}\right)' = (\gamma^2 + g)\, a, \qquad \mathcal{D}(x) = \gamma^2 + k_z^2 - u(x)^2, \qquad g(x) = \frac{2\alpha v^3 \Omega_A}{\gamma^2 + v^2\Omega_A^2}}$$



Physical picture

$\mathcal{D}$ changes sign exactly where the two branches meet: $\mathcal{D} > 0$ is a trapped shear well, $\mathcal{D} < 0$ is the tachyonic outer region — one mechanism, two fates. g is the only drive of the shear branch, and it dies as v^3 toward the layer center: the wave must shake the charges, make the two beams respond *differently*, and collect that difference as a current.

Exactness verified: the reduced equation reproduces the full six-field eigenvalues to better than 10^{-10} relative difference.

A growth-rate ceiling

The drive g is a resonance in the beam-frame precession frequency, maximized when that frequency matches the growth rate itself. Its peak value is $G_{\mathsf{max}} = \alpha V_0^2/\gamma$. A confining well needs $G_{\mathsf{max}} > \gamma^2$ somewhere, which forces

$$\boxed{\gamma^3 \leq \alpha V_0^2}$$

independent of k_z and of the shear-layer width.

Physical picture

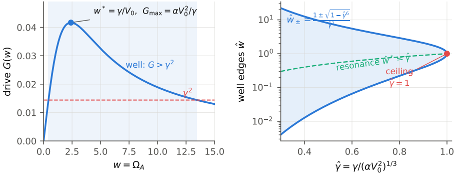
A self-limiting resonance: the beam response is a Lorentzian in local precession frequency, and its width is set by γ itself. The faster the mode grows, the broader the band it averages over, and the weaker the peak drive it can extract. The ceiling is the fixed point of that feedback.

Measured: peak growth rates reach 95–99% of $(\alpha V_0^2)^{1/3}$ in every simulated series tested (shown quantitatively later).

A universal, dimensionless well

In natural units $\Gamma \equiv (\alpha V_0^2)^{1/3}$ (rate scale) and $K \equiv (\alpha/V_0)^{1/3}$ (wavenumber scale), the drive becomes universal — no α, V_0 remain:

$$G(w) = \Gamma^2\, \hat{G}(\hat{w}), \qquad \hat{G}(\hat{w}) = \frac{2\hat{w}}{\hat{\gamma}^2 + \hat{w}^2}$$

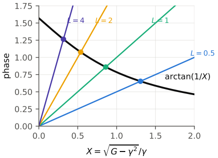


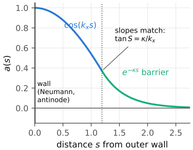
Growth rates scale as $(\alpha V_0^2)^{1/3}$ and wavenumbers as $(\alpha/V_0)^{1/3}$ — not as the naive product αV_0. This scaling is directly confirmed on GPU data later in the talk.

Quantization of the shear well

The well is shallow — less than a quarter wavelength of accumulated phase — so the growth rate is set by a connection condition at the well's edge rather than a simple cavity-mode count:

$$\tan S = \frac{\kappa}{k_{\mathrm{edge}}}, \qquad \kappa = \sqrt{\gamma^2 + k_z^2}, \qquad S = \int \sqrt{Q}\, dx$$





Physical picture

A surface-mode-like state, not a cavity mode: growth rates hug the ceiling (a small drop in γ buys all the depth needed), while the wavelength at which growth peaks is set by the geometry of the confining window rather than by the shear profile itself.

Solving this quantization condition exactly reproduces the independently computed six-field eigenvalues to $\leq 2\%$ (median $\leq 0.2\%$) across a wide survey of (α, V_0, k_z).

No intrinsic wavelength maximum

In an unbounded domain, nothing stops the mode from following the resonance outward as k_z grows: the peak drive $G_{\max} = \alpha V_0^2/\gamma$ does not depend on k_z, so $\gamma(k_z)$ climbs monotonically toward the ceiling — directly confirmed: γ reaches 99.5% of the ceiling by $k_z = 10$ in an unbounded test, with no turnover.

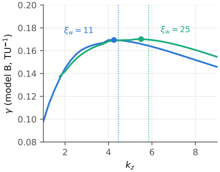
Physical picture

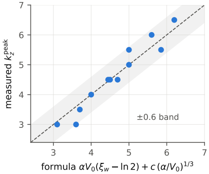
The peaked dispersion curve seen in the finite simulation domain is therefore **selected by the box, not intrinsic to the flow** — the single most counter-intuitive result of the theory, and the one most directly tested against data next.

The confinement-set wavelength

The resonance location moves outward linearly with k_z; setting it equal to the domain's confinement radius ξ_w gives the wavenumber at which growth peaks:

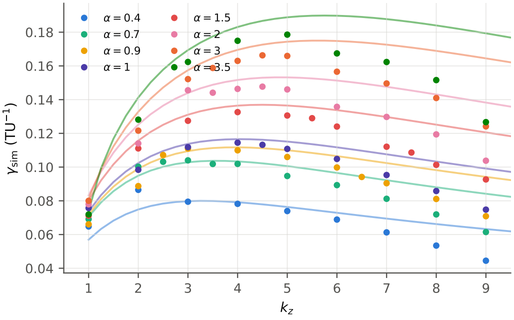
$$\boxed{k_z^{\mathrm{peak}} \simeq \alpha V_0(\xi_w - \ln 2) + c\,(\alpha/V_0)^{1/3}, \qquad c \approx 1}$$



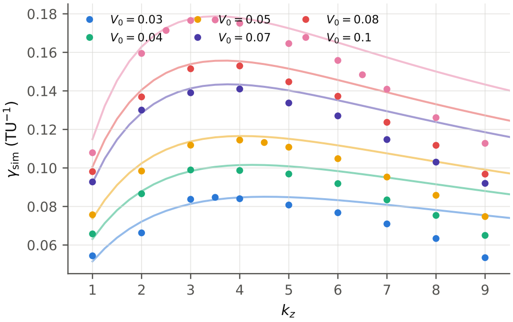


Both terms are independent of the shear-layer width — coupling strength and confinement radius jointly select the wavelength; the flow profile itself does not directly enter. This is the sharpest and most novel claim of the theory, and is tested directly on GPU data next.

Non-monotonic dispersion, on real simulation data



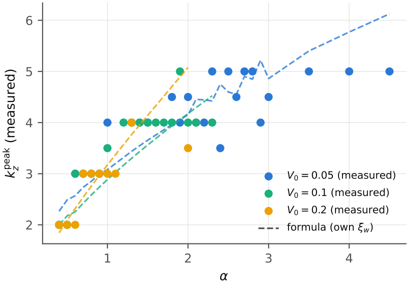
fixed $V_0 = 0.05$, varying α



fixed $\alpha = 1$, varying V_0

Points are simulated $\gamma(k_z)$; curves are the σ-chased eigensolver's theoretical prediction, drawn underneath. Every curve peaks at an intermediate k_z, and the peak location climbs with α — from $k_z^{\text{peak}} = 2$ at $\alpha = 0.4$ to 5 at $\alpha = 3.5$, at fixed shear width. Classical Kelvin–Helmholtz also peaks, but its peak is fixed by the shear width alone; a peak that moves with the gauge coupling, at fixed flow, has no classical analogue.

The peak tracks confinement, not the shear width



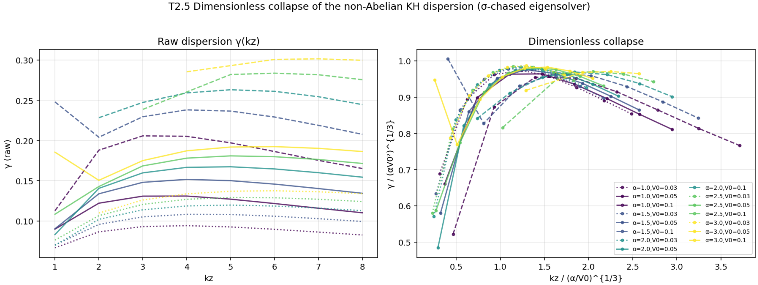
(α, V_0, ξ_w)	measured	formula
(1, 0.05, 20)	3.5	3.7
(2, 0.05, 11)	4.5	4.5
(3, 0.03, 5)	5.5	5.0
(5, 0.03, 5)	6.5	6.2
(1, 0.20, 10)	3.0	3.6

Direct window test, same (α, V_0): widening the confinement radius alone moves the measured peak from $k_z = 4.5$ to 5.5, matching the formula's $4.45 \to 5.85$.

Physical picture

Two different confinement radii, same coupling and shear velocity, two different — and both correctly predicted — peak wavenumbers: confinement and coupling jointly select the wavelength, not the flow profile.

The dimensionless collapse

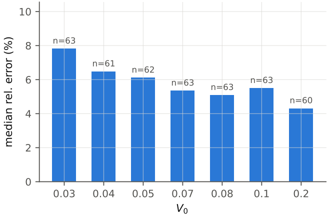


$$\gamma_{\mathrm{peak}}/(\alpha V_0^2)^{1/3} = 0.977 \pm 0.011 \quad (1.1\% \text{ spread, } 10\times \text{ range in } \alpha V_0)$$

The single most quantitative confirmation: fifteen independent dispersion curves, spanning a factor of ten in αV_0, collapse onto one master curve — direct proof that α, V_0 enter growth rates through $(\alpha V_0^2)^{1/3}$, exactly as the ceiling predicts.

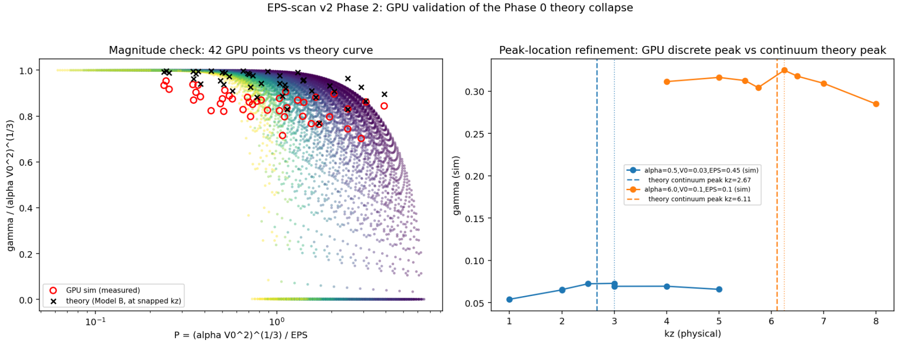
Full-map accuracy

Growth rates measured across a broad grid in α, V_0, and k_z (1087/1134 points, 96% coverage) against the independently computed eigenvalue, at fixed shear width:



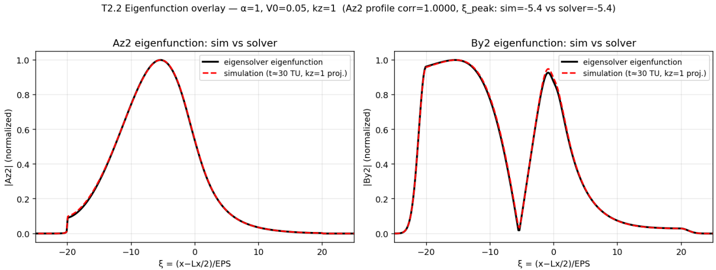
Median relative error **4–8%** depending on V_0, improving monotonically with higher shear velocity (int-tier points, $\alpha \in [0.4, 1.0]$ common to every V_0 so the bars are an apples-to-apples comparison rather than an artifact of which α's each campaign happened to sweep). The theory and the simulation agree across the whole mapped parameter space, not just at isolated anchor points.

Independence from the shear-layer width



The full ceiling-and-peak theory, re-expressed in terms of a single dimensionless group combining coupling, shear velocity, and layer width, collapses 91 simulation runs spanning a wide range of shear-layer widths onto one universal curve: median relative error 9.9%. The wavelength-selection law does not depend on the microscopic shear width — only on coupling and confinement.

The eigenfunction structure matches, not just the rate



Spatial profile of the dominant field, simulation vs. theory: correlation $= 1.0000$. A companion field, seeded at less than 1% of the driving field's amplitude and grown entirely through the instability's internal feedback chain, independently reproduces the theory's non-trivial double-lobe shape — an emergent structural match, not just a matched growth rate.

Where this leaves the physics

Result	Status
$k_z = 0$ chromo-Weibel rate	Closed form, validated to 0.5%
Six-field linear system	Validated to 1–9% against simulation
Exact scalar reduction	Reproduces linear theory to $< 10^{-10}$
Growth ceiling $\gamma^3 \leq \alpha V_0^2$	Measured at 95–99% of ceiling; 0.977 ± 0.011 collapse
Quantization / dispersion curve	Matches theory to $\leq 2\%$ (median $\leq 0.2\%$)
Confinement-set wavelength selection	Confirmed directly by window tests and full parameter map

A shear-driven non-Abelian instability, derived exactly and confirmed on over a thousand GPU simulation runs: coupling and confinement jointly select the wavelength; the flow profile does not enter directly.

Context and next steps

- No prior literature treats a genuinely shear-driven non-Abelian instability with a full wavenumber map — this appears to be new physics, not a variant of a known result.
- Closest related work: chromo-Weibel instabilities (anisotropy-driven, not shear); the outer tachyonic branch is a Nielsen–Olesen-type mode; Abelian shear instabilities (Das & Kaw; Csernai et al. in heavy-ion collisions) are the closest classical analogue but lack the non-Abelian wavelength-selection mechanism.
- Remaining work: a citation-trail audit before submission, and a dedicated follow-up study of the high-coupling regime, which is currently kept outside the headline claims while its numerical behavior is characterized more fully.

The theory is exact, matches simulation to a few percent, and is confirmed across a broad parameter survey.

Thank you.

Questions, and where to push next.